

Version II

*One profile, personalized from the phone,
working at every Caffevolve machine*

Prepared for Brent Langevin
October 2026
Working draft · Confidential

This plan is separate from the Caffevolve™ Platform Reference, which documents what has been developed so far. It contains unpublished concepts and should not be shared outside the company before the company's patent counsel has been consulted.

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1. Purpose and status

This plan describes **Version II** of the Caffeolve™ Platform. It is a separate document from the Caffeolve™ Platform Reference, which records what has been developed so far. Nothing in this plan is described in the Reference as already built, and nothing in the Reference has been changed to anticipate it.

The plan is a working draft. Each theme in section 2 carries one of three labels: **Agreed**, meaning its structure was settled in review; **Concept**, meaning a direction that has been explored but whose details are not settled; and **Planned**, meaning intended for an upcoming version after the core of Version II. Detailed policies and edge cases are deliberately left for later drafts (section 16).

Confidentiality. This document contains unpublished concepts. Before it is shared outside the company, or before any public disclosure of the features it describes, the company's patent counsel should be consulted.

2. Version II at a glance

A person sets up one profile, once. From then on it goes with them: the machine at home, the machine at the office, and the machine at a café that has opted in. At each one they identify themselves, the machine already knows their caffeine standing, their plan for the day and any Sub-Profile window that is open, and they press brew. The personalizing that makes this possible is done on the phone, not standing at the machine.

Theme	What it adds	Status
Personalize from the phone	CaffeolveGo becomes the authoring surface. Accounts are created by an emailed invite. Profile extras, Sub-Profiles and Routines are created in the app. The profile and the CBI can be completed at the device or in the app.	Agreed
One profile, any of my devices	One account holds one profile, used at home and at an equipped office. A second device attaches the existing profile with no new CBI.	Agreed
Caffeolve Away: the visitor pass	A short-lived permission, granted from the app, that lets an opted-in machine such as a café's use the profile for one visit. Regular stops use a standing pass.	Concept
Paying from the app	The app shows the total price and the user approves it before any payment starts, so there is no queue at the till.	Planned

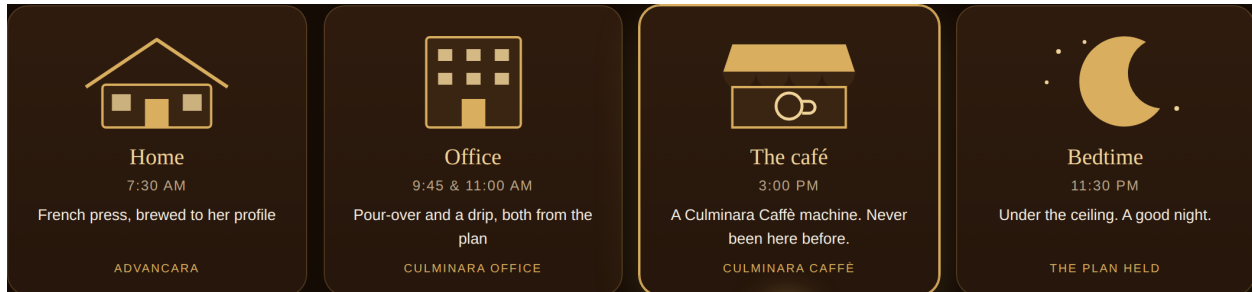


Figure 1. Illustration: one profile across a day, at home, at the office and at a café. Figures are illustrative.

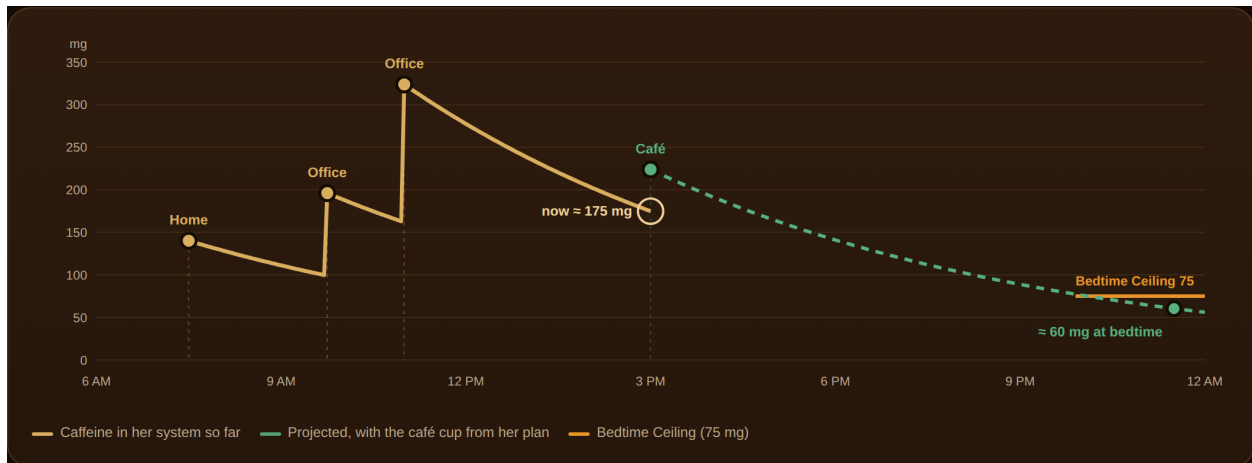


Figure 2. Illustration: Diane's caffeine level across the day, with the café cup from her plan. The Bedtime Ceiling holds.

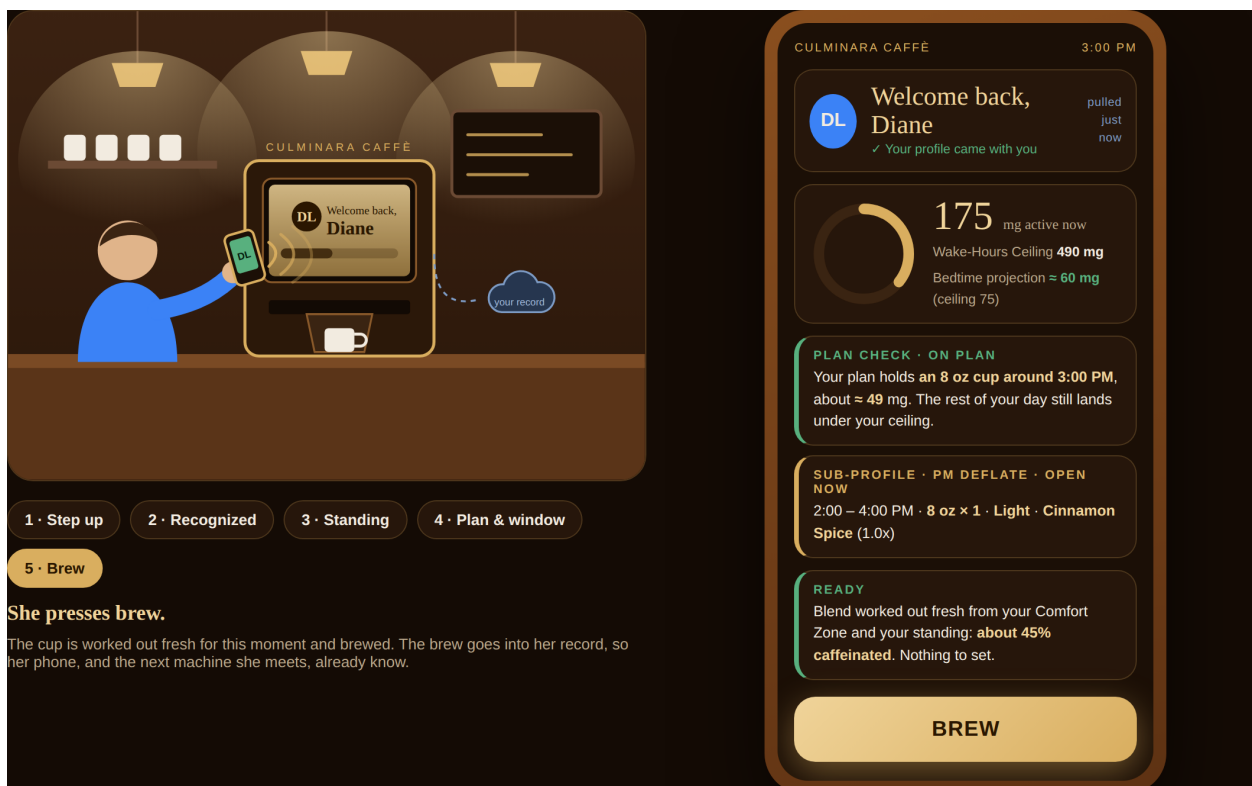


Figure 3. Illustration: at a Culminara Caffè machine the profile is recognized, the standing, the plan and the open Sub-Profile window are shown, and the visitor brews.

3. Design principles

1. **The phone is the authoring surface; the device is the execution surface.** All brew, grind and drink-path decisions, and every brew initiation, stay at the device. Personalizing can be done wherever the user is comfortable.
2. **The platform advises; the user decides.** This holds for ceilings, plans and payment alike.
3. **The device works without the cloud.** It brews from its own copy of the profile and plans. Connectivity adds completeness, never capability.
4. **The cloud is a store, not a brain.** It carries a person's record between their devices. It does not compute residual or decide what to brew.
5. **One account, one profile, permanently.** A profile is bound to its account and is never moved to another.
6. **The administrator decides who holds a slot; the user completes everything personal.** The administrator never answers the Caffeine Behavior Instrument for anyone.
7. **Keep only what is needed.** The email address is the account ID, no phone number is kept, Caffeolve stores no card details, and a machine drops a visitor's data when the visit ends.

4. The journey: device set-up to a finished profile

FIG. 44A–D (Appendix A) draw this flow in full, with the administrator's steps, the user's steps and the platform's own steps shaded differently. Each step below is marked **Existing**, meaning it is already documented in the Platform Reference or built in the current prototype, or **New**, meaning it is introduced by Version II.

1. **Existing. The administrator sets up the device:** Administrator PIN, language and region, Wi-Fi, beans, and additives with their canister slots.
2. **Existing. A resident profile is reserved by the administrator** behind the Admin PIN, who enters the first name, initials (2–3 characters) and profile color, which reserves one of the device's slots (Originara 2, Advancara 5, Culminara Office 20). **New.** The administrator also enters the user's **email address** (the Basic Resident Profile), and the administrator's own profile is reserved the same way as everyone else's.
3. **New. The device registers the shell** (name and email only) with the cloud, which emails a one-time invite code and a link to the app. The user's personal data stays on the device.
4. **Existing. The profile appears as “Ready for you.”** The resident brewing paths stay locked until setup is finished, and Guest Brew is available meanwhile. The user can pause setup and resume it, but cannot discard the slot.
5. **New. The user redeems the code in the app**, which creates the account. The invited email address is the user ID, verified by the redemption itself, and the user sets a password.

6. **Existing. Profile setup:** the Caffeine Behavior Instrument, the Caffeine Comfort Zone it produces, and the option to fine-tune the two ceilings, at the device. **New.** It can also be completed in the app.
7. **Existing. The profile extras** (additive preferences, grind setting, Sub-Profiles, the BrewNow! Profile, workplace settings) exist today at the device. **New.** In Version II they are created in the app, with defaults of no additives and a medium grind until they are set.
8. **New. Plans and Routines** change as described in section 9.

The Guest path is unchanged: it stays entirely at the device, with no account, no invite and no CBI.

5. Accounts, invites and pairing

- **The invite.** The code is single-use, expires after a short window, locks after a few wrong attempts, and binds to the first account that redeems it. The administrator can issue a fresh one, correcting the address if needed.
- **The account.** The user ID is the invited email address. A password reset goes by email, with no administrator involved. A new or second phone is handled by signing in with the same email and password.
- **One account, one profile, bound permanently.** There is no reassign function. If the wrong person claims a profile, for example because the address was mistyped, the administrator removes the profile, which also deletes the cloud copy and unpairs the account, and reserves a new one.
- **Cleanup.** If a code expires unused, the shell's email address is deleted from the cloud. The platform holds no phone number.
- **An unclaimed profile works fully at the device,** but exists only there, with no cloud backup. The device and the app say so when the invite is offered.
- **Phone-first.** A person cannot self-create a resident profile on a household device. The administrator-issued invite is the only route to a slot.
- **The Administrator PIN** secures the device and decides who holds a profile on it: device configuration, reserving a slot, issuing an invite, removing a resident profile, and resetting or transferring the device.

6. What is created where

A resident profile is built in two stages by two different people: the **Basic Resident Profile** (the administrator's reservation) and **Profile setup** (the user finishing it), which produces the **Complete Resident Profile**. Today both stages happen at the device. The last column shows what Version II changes.

Item	Create	Edit	In Version II
Basic Resident Profile: name, initials, color	Administrator, at the device (Admin PIN)	As today	Unchanged; the email address is added

Item	Create	Edit	In Version II
Complete Resident Profile: the CBI and the ceilings it derives	User, at the device or in the app	Either surface, through the profile review screen	New: can also be done in the app
Profile extras: additive preferences, grind setting, Sub-Profiles, BrewNow! Profile, workplace settings	App only	Either surface	Changed: creation moves from the device to the app
Today's plan (Plan my Day)	Either surface; lasts that day and is not saved	Either surface	New: also in the app
Routines (saved plans)	App only	App only	Changed: saving moves from the device to the app
Guest profile	Device only	Device only	Unchanged

- **New: the device is never unusable for want of a phone.** A resident can finish the CBI and ceilings at the device with no phone and no account.
- **New: resumable across surfaces.** The CBI can be paused with “Exit setup and complete later” and finished on either surface. The profile is established only when the CBI completes; until then the device uses conservative guest-style defaults.
- **Existing:** bedtime is not part of the profile. It is a session value set and confirmed at the device with each brew.
- **New: on the device,** the profile setup screens for the profile extras say they are available in the CaffeolveGo app.

7. Ceilings, learning and Edit Profile

Existing. The Reference describes the Caffeine Comfort Zone (Peak Zone Floor, Wake Hours Ceiling, Bedtime Ceiling) and the Caffeine Decay Rate as derived from the CBI and refined over time by the Adaptive Feedback System. The prototype lets a user fine-tune the two ceilings from the profile review screen, with a “Use computed” option that removes the adjustment, and an advisory warning, never a block, for a high setting. Version II keeps all of this and adds the following.

- **New: the model is stated.** Each ceiling has three parts: the computed value (from the CBI, moved by the Adaptive Feedback System), an optional user adjustment, and the effective value, which is the adjustment if one exists and otherwise the computed value. The device brews against the effective value.
- **New: only the two ceilings are adjustable.** The Peak-Zone Floor and the Caffeine Decay Rate are not user-adjustable. The Reference’s Phase 2 review text invites the user to adjust any boundary value; this will need to be reconciled (section 14).

- **New: the Adaptive Feedback System never overwrites an adjustment.** It keeps refining the computed value in the background, so “Use computed” returns the learned value, and its boundary-triggered prompts fire around the effective value.
- **New: refinement is stored as feedback events,** each immutable, with a source and an ID. The computed value is derived from the baseline plus those events, so events from either surface merge without conflict.
- **New: Edit Profile has a stated reset rule.** Edit Profile re-runs the whole setup from the profile review screen, on either surface. If the CBI answers are unchanged, only the other fields change and nothing is reset. If the CBI answers change, a new baseline begins: the computed boundaries and decay rate are recomputed, any ceiling adjustments are cleared, and the Adaptive Feedback System starts over.

8. Staying in step between phone and device

- **Every edit is a new version** of the record, stamped with its source (phone or device), time and the version it was based on.
- **The later edit wins field by field.** A ceiling change on one side and a rename on the other both survive. If both changed the same field, the later one wins and the other surface shows a one-line notice.
- **The two ceilings are treated as a pair,** because they are adjusted together, so the result is never a mix the user did not see.
- **A re-run of setup that changes the CBI answers** starts a new baseline, and an edit made against the old baseline on the other surface is superseded, with a notice.
- **The device holds a complete local copy** of the active profiles and plans and keeps brewing from it if the network or the app is unavailable. Each side shows when it last synced.
- **Brew events** remain device-authored and the device remains the source of truth for them. For personalization records, the most recent valid version is authoritative.

9. Plan my Day: what changes

Unchanged. Plan my Day is documented in Reference Section 5 and walked through in Section 8.4 and FIG. 43: the whole-day projection against both ceilings, shaping the day, protected drinks, Rebalance, Routines and one-off days, Plan Check, and the rule that a different bedtime declared mid-day re-solves the whole day. Version II changes none of that. It changes three things.

1. **Routines are managed in the app only.** A Routine is created, edited, named, given its *saved days* and deleted in the app. The device never saves or changes a Routine. The device can still build and adjust today’s plan (add, change or remove a drink, protect one, run Rebalance, or skip the plan), and that lasts that day only. Today, saving happens at the device-side Planner.
2. **Two bedtimes, with a stated precedence.** The *planned bedtime* is a required entry when a plan is set up and lives with the plan, not the profile. The *session bedtime* is set and confirmed at the

device when a brew begins, and it always rules. It applies to that day only and never writes back to a Routine. The Reference does not draw this distinction.

3. **A bedtime declared in CaffevolveGo counts the same way as one declared at the device,** because the app is an extension of the device. The most recent declaration applies.

10. One profile, more than one device

- One account holds one profile, and it works on every Caffevolve device the person uses, at home and at an equipped office.
- **Joining a second device (FIG. 44D).** The second device’s administrator reserves a slot with the person’s name, initials, color and the email address of their existing account, and issues the usual invite, which must match the account’s email. The person redeems it while signed in. The existing profile attaches to that device with no new account, no second profile and no new CBI. Each device an account uses consumes one slot on that device.
- **Name, initials and color are per-device display settings.** They are not identifiers, are set by each device’s administrator, are not synced between devices, and may differ between home and office. Initials are 2–3 characters, and a color already used on a device is unavailable, so every profile on a device stays distinct.
- **The profile PIN** exists only for the shared Culminara Office device. It is a user-chosen four-digit PIN required when a profile is selected on an Office device, never at home, and never for changes made in the app, which are made in the user’s account with email and password.

11. Caffevolve Away: the visitor pass

Working name. Status: Concept. The visitor pass would let a person use their one profile at any Caffevolve machine that has opted in, such as a café’s, with no profile slot, no invite and no account on the machine. FIG. 45A shows a first visit and FIG. 45B a regular stop.

- **The venue opts in.** The machine’s administrator turns on “Allow visitors” in Administrator Functions, behind the Admin PIN. Machines that have not opted in work as they do today.
- **The visitor grants a pass.** At the machine the visitor scans or taps with the app. The app names the venue and what is shared, and shows the total price where the venue charges, so one approval covers both. The phone is the key, so no PIN is typed on a public machine.
- **A short-lived pass.** The cloud issues a pass bound to that one machine. It ends when the visitor walks away, after about two hours at most, or when the visitor ends it in the app.
- **Who knows what.** The machine shows a one-time code. The app answers it with the visitor’s signed-in account, and the cloud checks whether the visitor has a standing pass for that machine. With none, the app asks for consent and the cloud issues a one-visit pass. With one, a single tap confirms presence and the cloud issues the token. Either way the machine receives only a signed visit token saying what it covers and when it ends, so it behaves the same. The standing pass lives in the visitor’s account, not on the machine.

- **The machine borrows the profile.** It pulls the profile and today's caffeine record, recomputes the standing with the same engine as every other device, and shows Plan Check and any open Sub-Profile. When the visit ends it deletes its copy.
- **What a visitor can do.** Brew, and adjust, protect or skip today's plan. A visitor cannot change the profile. The machine never sees the email, the account or the CBI answers. The brew goes into the visitor's own record, and the venue receives no personal data.
- **Regular stops.** For a place visited several times a week, the app offers to make it a regular stop. A standing pass is saved for that machine, so there is no daily pass and no consent screen. Each visit starts with one tap to confirm presence (where the venue charges, that tap is the approval of the total price), and the machine pulls everything fresh and keeps no profile data between visits. The standing pass ends when the user removes the place in the app, or lapses after about 90 days unused.
- **In the app,** a "Places" list shows the regular stops, any visit in progress with an "end this visit" button, and a short visit history.
- **The machine adapts to itself.** A café machine has its own beans, additives and grind range. An additive in the visitor's preferences that the machine does not carry is skipped, and the visitor is told. The blend is worked out for that machine's beans.
- **Bedtime and measurement.** Bedtime is confirmed at the machine for each brew and can be pre-filled from the app. A café brew is a measured entry, because the machine knows the exact dose.
- **One visitor at a time per machine.** Without the app or a connection, the Guest path remains.
- **Not the same as joining a second device.** That is membership with a slot for a device the person uses regularly. The visitor pass is a guest arrangement for a machine the person does not belong to.

12. Paying from the app

Status: Planned for an upcoming version. A café's speed is lost if a visitor must queue to pay. Payment therefore belongs to the same walk-up experience as the pass. FIG. 46A shows the flow and FIG. 46B how it is set up in the app.

- **The app always shows the total price,** including any tax or extras, and the visitor approves it before any payment starts. There is no automatic payment and no spending limit.
- **One approval, not two.** On a first visit it is combined with allowing the pass. At a regular stop it doubles as the presence tap. Where a venue does not charge, the plain presence tap remains.
- **Approval is by the phone,** with its fingerprint, face or phone PIN. If the total changes after approval, a new approval is needed for the new total.
- **Caffeolve stores no card details.** Payment goes through the phone's own wallet or an established payment provider, and the money goes to the venue's own merchant account.
- **The machine brews only after payment is approved.** If the brew fails, the payment is voided or refunded automatically. A receipt goes to the app, and the venue sees the sale, not the visitor's

profile.

- **At venues without a Caffeolve machine**, paying from the app for what the user ordered, with no link to the venue's ordering system, is a possible later extension under the same approval rule. How the venue would be paid is left open.

13. Security, access and data governance

- **The Administrator PIN** remains the single gate for device configuration and the roster of users. **The profile PIN** applies to Office devices only. Profile edits at a home device, and all edits made in the app, need no further PIN.
- **The user's email address and password** secure the account. Password rules, lockout and any second sign-in factor are to be specified (section 16).
- **CBI answers describe sleep and caffeine habits** and are health-adjacent, and an email address entered by the administrator is personal data about another person. Consent, retention, export and account deletion rules must be set before launch, in addition to per-user isolation and encryption in transit and at rest.
- **Removal deletes**. When the administrator removes a profile, the cloud copy is deleted and the account is unpaired.
- **Version matching extends** to the record schema and the shared personalization flows, so firmware and app do not drift apart.
- **The assumption should be tested**. Completion rates for the CBI and planning on a phone prototype should be compared with the device before the full build.

14. Effect on the Platform Reference

The Platform Reference is unchanged and describes only what has been developed. When the features in this plan are built, the following passages of the Reference will need to change. They are listed here so that nothing discussed is lost.

Reference section	Change when Version II ships
1. The Caffeolve™ Platform	Mention the app as a personalization surface, and note the later extensions to other machines and to payment.
4.1 Resident and Guest Profiles	The administrator also enters the email address; the invite and account; the CBI at the device or in the app; profile extras created in the app; one account holding one profile used on every device; per-device display settings.
4.2 Sub-Profiles and the BrewNow! Profile	Created in the app and edited on either surface.
3. The Adaptive Feedback System (Phase 2 review)	The review invites the user to adjust any boundary value; Version II allows adjusting only the two ceilings. Reconcile the text.

Reference section	Change when Version II ships
4.3 Profile Security	The profile PIN applies only on an Office device, including for a profile shared with a home device. Changes made in the app use email and password.
5.3 Routines and one-off days	Routines created and managed in the app with saved days; today's plan adjustable on either surface and not saved.
5.5 When the real day differs from the plan	Planned bedtime and session bedtime; the plan re-projected whenever a session bedtime is set.
7. CaffeolveGo	The app as a personalization surface; the visitor pass and payment as later sections.
8.1 Set-up screens	The first profile is created by the administrator like any other; the email field and the invite step; account creation; profile extras moved to the app.
8.4 Plan my Day	Routines saved from the app, with saved days (the walkthrough currently shows saving on the device-side Planner).
FIG. 43A	Step 3802: profile load excludes bedtime; planned bedtime entered in the workspace; “assign weekdays” (3832) and “matching the current weekday” (4106) become saved days; add a phone authoring entry point.
Contents and Index	New entries for the account, the invite code, saved days and the visitor pass; TOC page numbers renumbered.
Device tiers	Define the Culminara Caffè tier, which appears in the prototype but not in the Reference.

15. Development roadmap

Prototype status

Done in prototype v2. An email address field (required, validated, unique per device) on the administrator's “Set Up a New User” step and on the first-profile step, hidden on Edit Profile, and an on-screen confirmation that an invite code was emailed. The prototype deliberately still lets a whole resident profile, extras included, be set up at the device, so that the full profile remains available for demonstrations.

Remaining work to complete Version II

- **App-side screens:** redeem the invite code, create the account (email as user ID, verification, password), sign in, reset a password by email, and the profile setup and review screens.
- **Join another device:** an existing account attaches its profile to a second device.
- **User profile setup flow at the device** (not device setup): the profile extras created in the app, with “available in the CaffeolveGo app” at the device and the defaults of no additives and a medium

grind. Editing existing extras stays available on either surface.

- **Plan my Day:** today's plan on either surface, Routines in the app only, the planned bedtime, and re-projection when a session bedtime is set.
- **Platform behavior:** issuing and redeeming invite codes, registering the shell with the cloud, account binding, versioned per-field sync with the ceilings as a pair, feedback stored as events, and the device's full local copy.
- **Later themes:** the visitor pass (Allow visitors setting, pass issuing and expiry, scan or tap, regular stops, Places list) and payment from the app.

Proposed sequencing (for confirmation)

Phase	Scope
1. Personalization and accounts	Sections 4 to 10: invites and accounts, profile extras and Routines in the app, one profile on more than one device, sync and local copy.
2. Caffevolve Away	Section 11: the visitor pass and regular stops at opted-in machines.
3. Payment from the app	Section 12: total-and-approve payment at Caffevolve machines, with the venue case left open.

16. Deferred decisions and open questions

These are left for later drafts. None changes the structure above.

- Password rules, the lockout policy and any second sign-in factor.
- Device resale and transfer: resetting a device, detaching it from accounts, and a “My devices” list in the app.
- Invite delivery details: bounced or filtered messages and the length of the expiry window.
- Whether a later session bedtime should lead the planner to offer restoring strength to drinks eased by an earlier Rebalance.
- How a venue without a Caffevolve machine would be paid under the later payment extension.
- The definition of the Culminara Caffè tier, including whether it has resident slots at all.
- AFS learning state, sync and pairing are specified here but not yet modeled in the prototype.

Appendix A. Figures

FIG. 44A–D follow the flow from device set-up to a finished profile. FIG. 45A–B draw the visitor pass, and FIG. 46A–B payment from the app. Method block numbers (4200s to 4900s) continue the series used for FIG. 43. Each figure also exists as a separate file in the figures folder.

CAFFEVELOVE™ · UTILITY METHOD SET · DEVICE SET-UP TO PROFILE SET-UP (NEXT VERSION, DRAFT FOR REVIEW)

FIG. 44A — Device Set-up · method block 4200s

DEVICE **APP** **DEVICE OR APP** **CLOUD** **NEW** = not yet in the current UX prototype

ADMINISTRATOR **USER** **AUTOMATIC** = who acts: shading shows the administrator, the user, or the platform itself

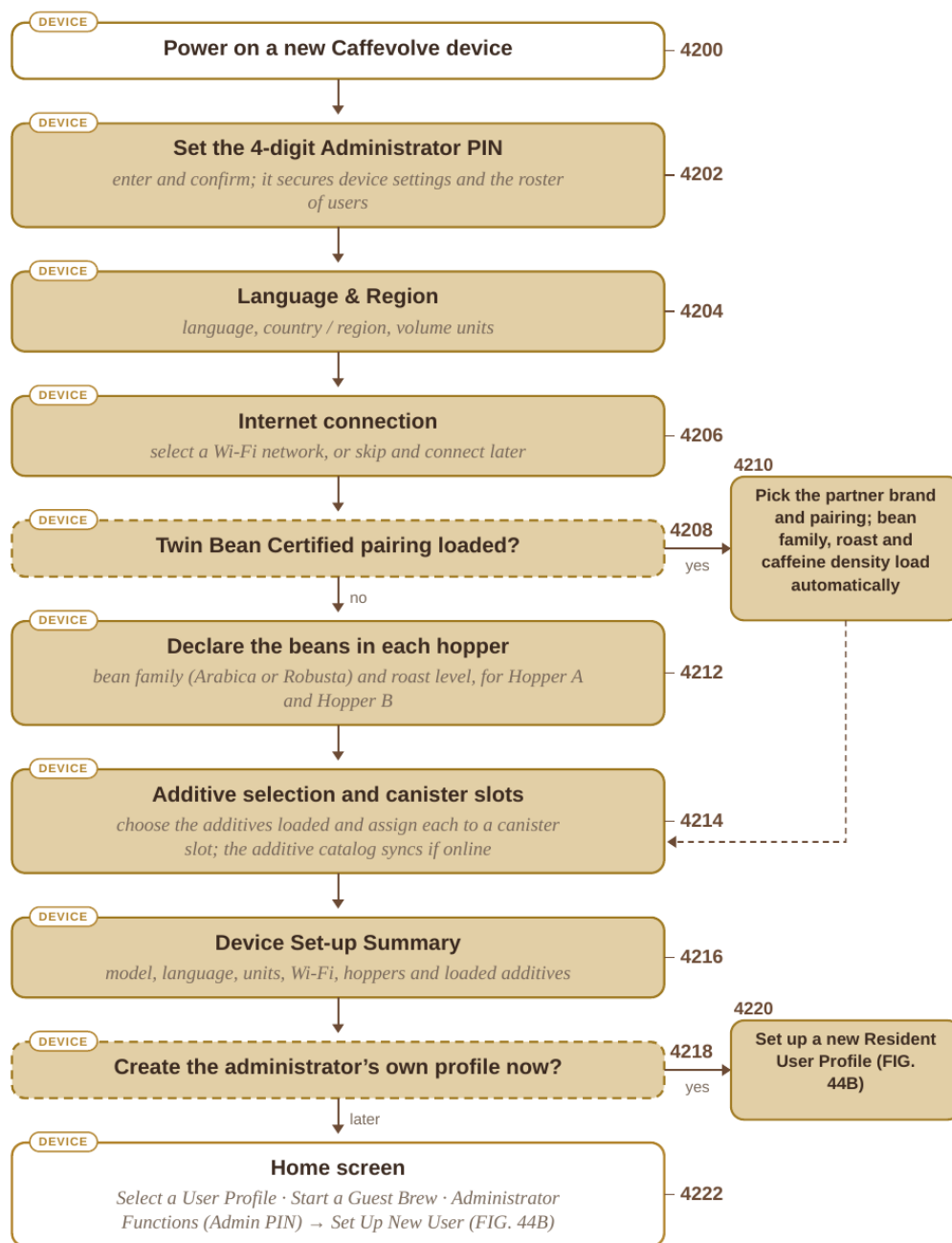


FIG. 44A. Device set-up.

FIG. 44B — Basic Resident Profile, Invite and Account · method block 4300s

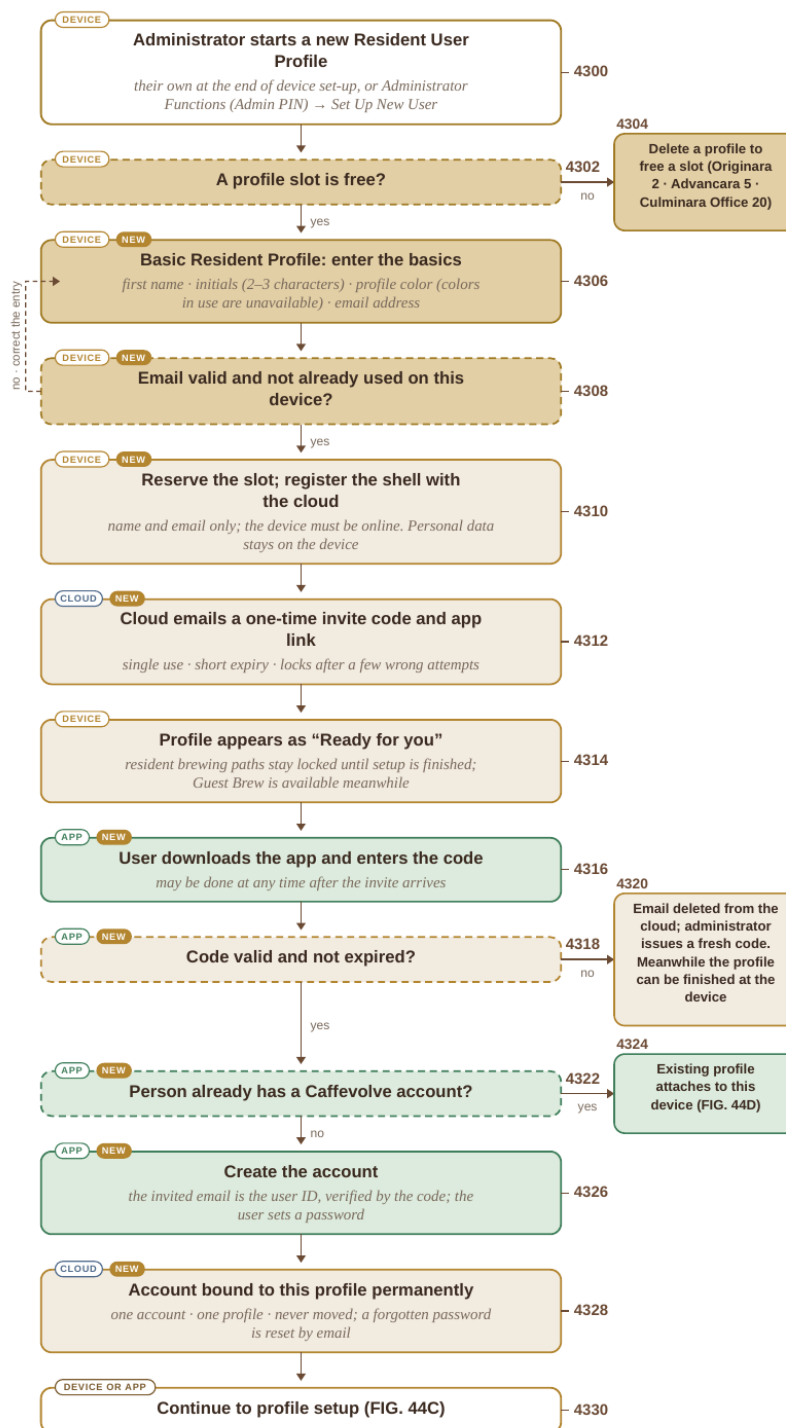


FIG. 44B. Basic Resident Profile, invite and account.

FIG. 44C — Profile Setup and Review · method block 4400s

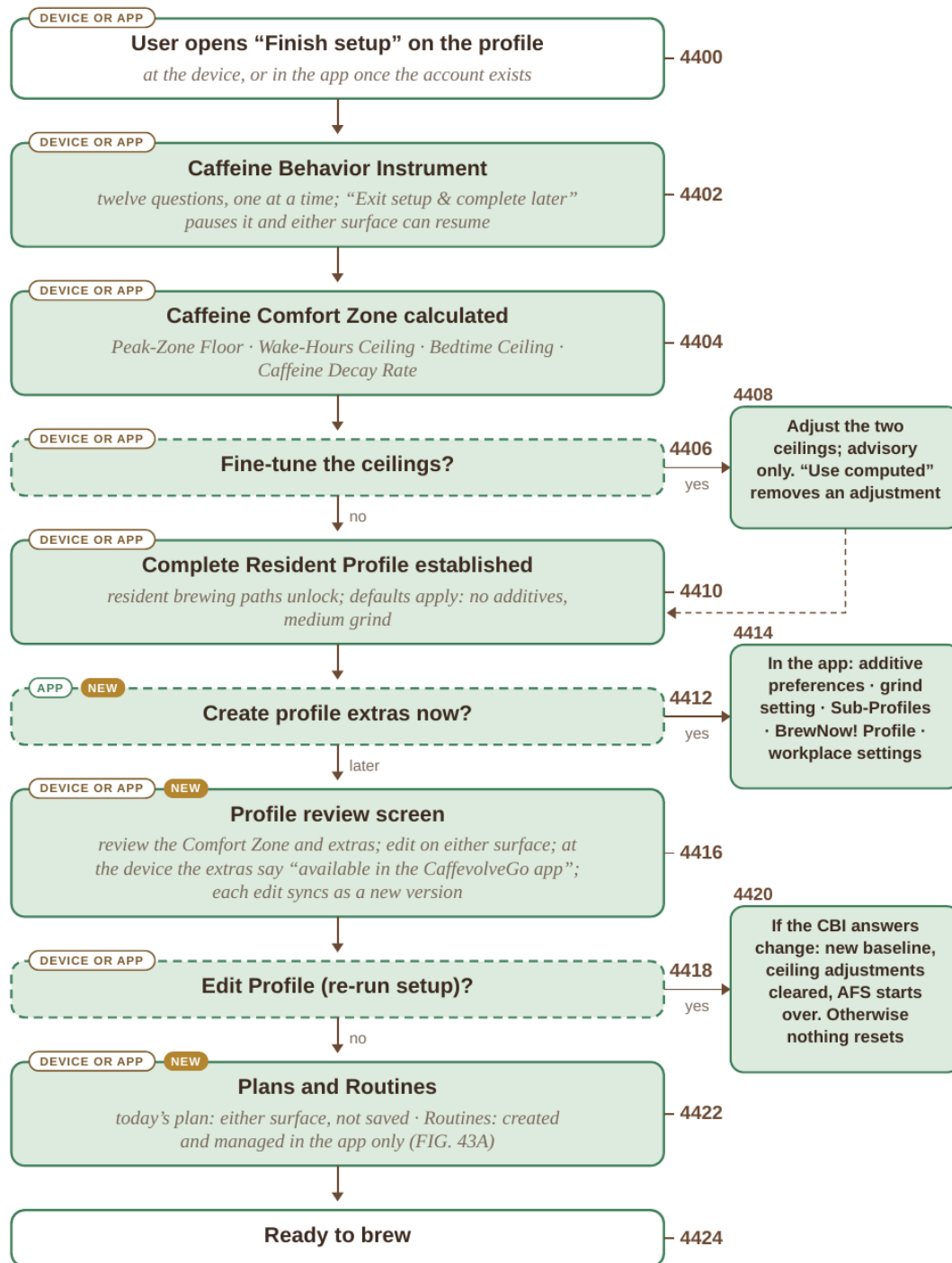


FIG. 44C. Profile setup and review.

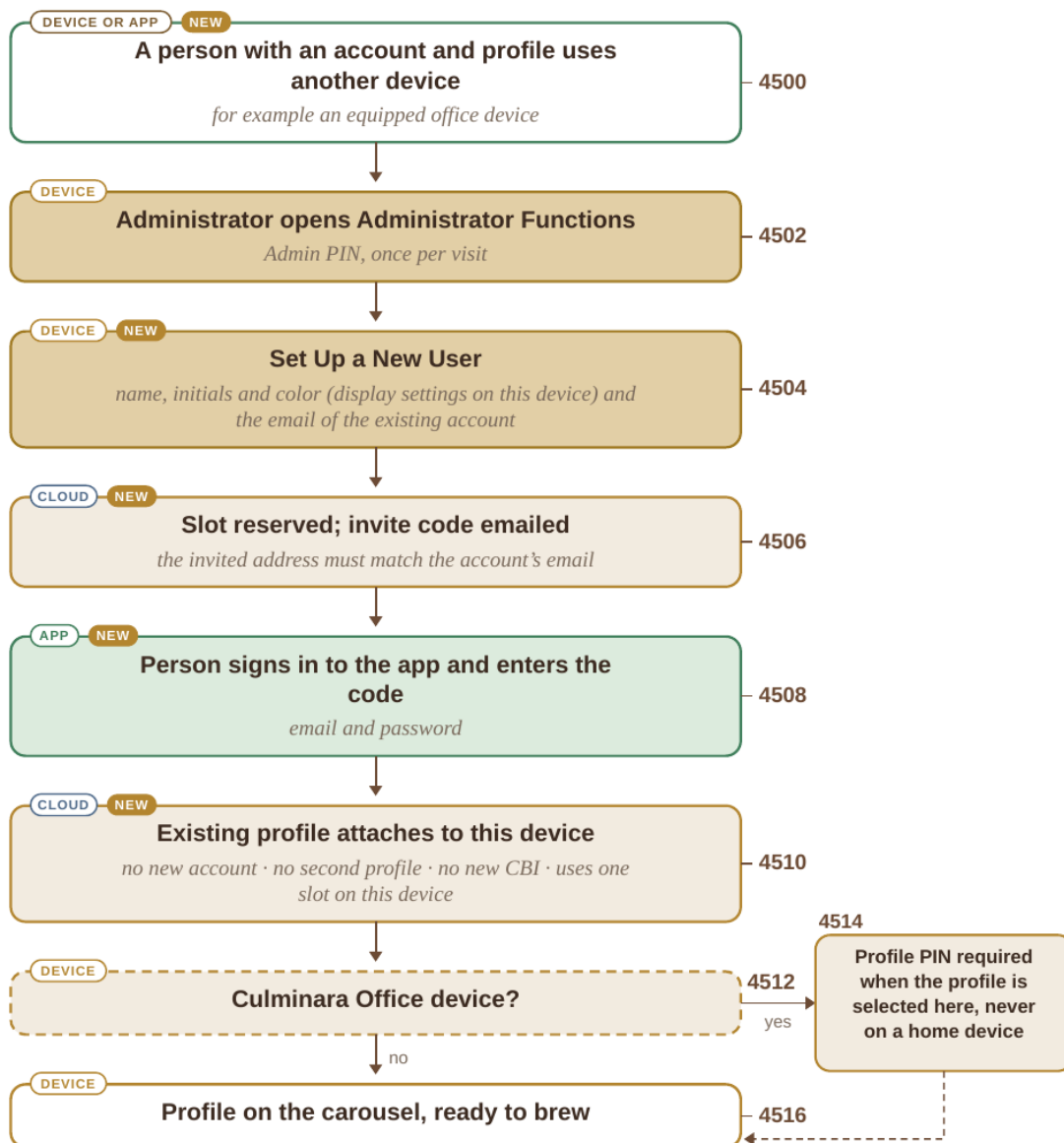
FIG. 44D — Joining a Second Device · method block 4500s

FIG. 44D. Joining a second device.

CAFFEOLVE™ · UTILITY METHOD SET · VISITOR PASS (FUTURE CONCEPT, DRAFT FOR REVIEW)

FIG. 45A — Visitor Pass: A First Visit · method block 4600s

DEVICE APP CLOUD Future concept: not part of the current design or prototype

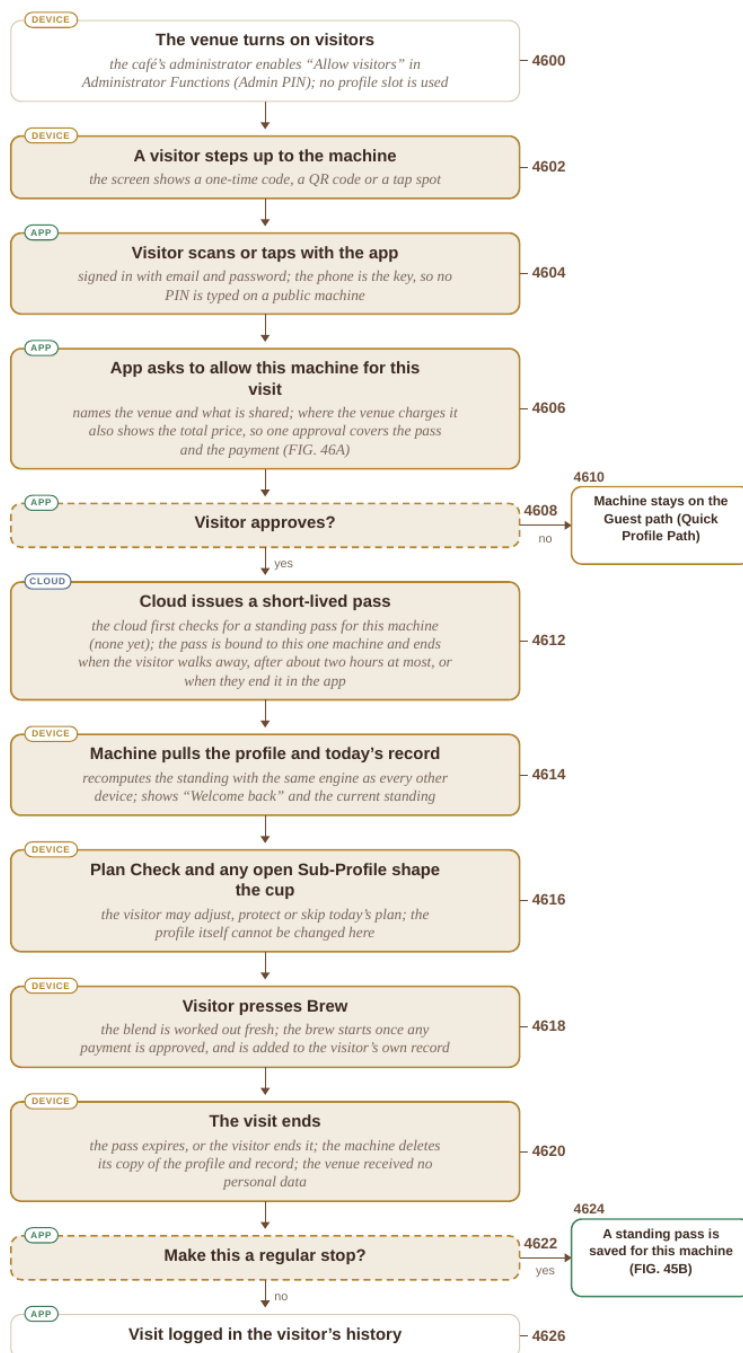


FIG. 45A. Visitor pass: a first visit.

FIG. 45B — Visitor Pass: A Regular Stop · method block 4700s

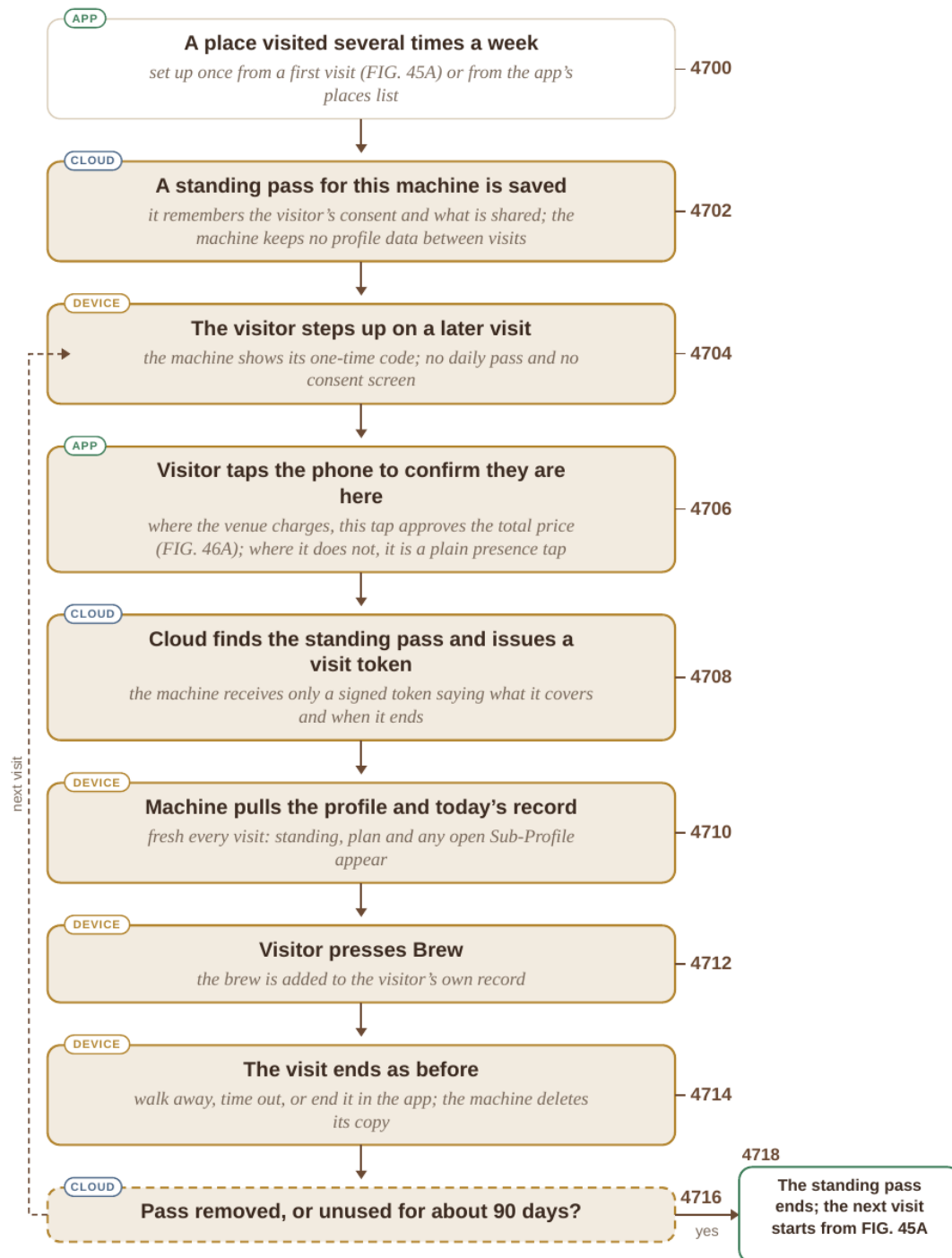


FIG. 45B. Visitor pass: a regular stop.

CAFFEOLVE™ · UTILITY METHOD SET · PAYING FROM THE APP (PLANNED, DRAFT FOR REVIEW)

FIG. 46A — Paying for the Cup · method block 4800s

DEVICE
APP
CLOUD
PAYMENT PROVIDER
 Planned for an upcoming version: not part of the current design or prototype

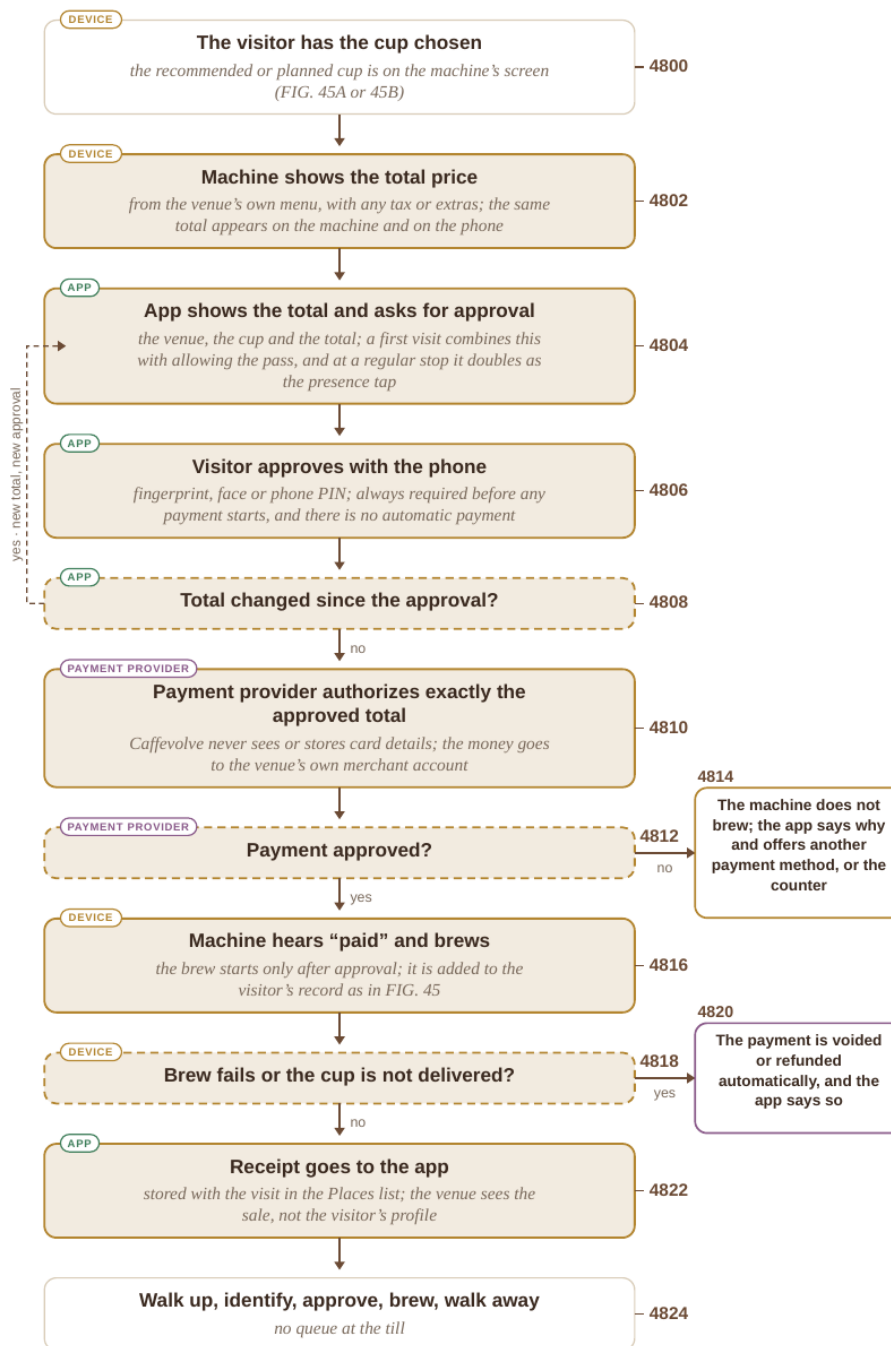


FIG. 46A. Paying for the cup.

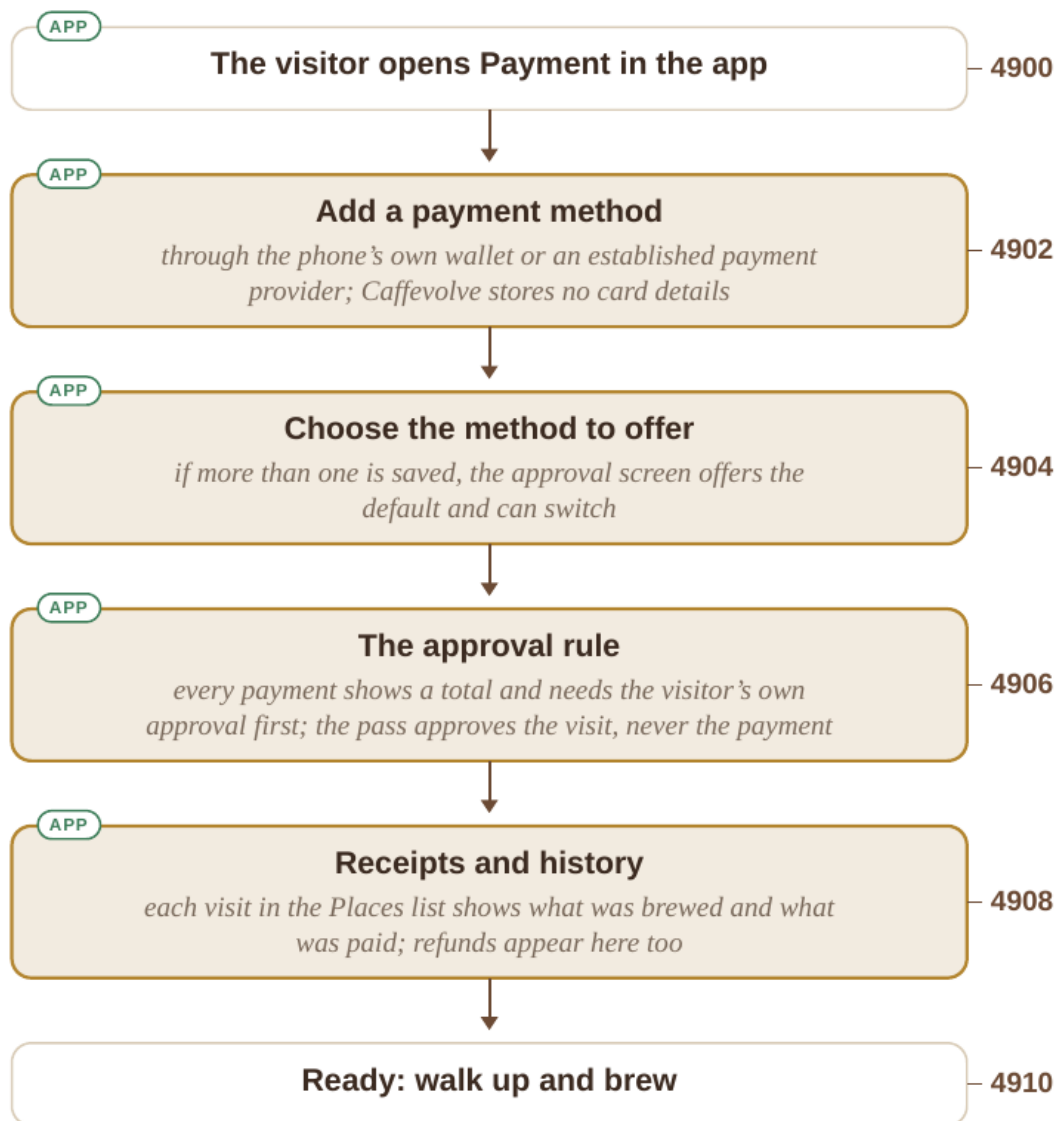
FIG. 46B — Setting Up Payment in the App · method block 4900s

FIG. 46B. Setting up payment in the app.

Appendix B. Glossary

Term	Meaning
Basic Resident Profile	The administrator's reservation of a slot: first name, initials, color and the user's email address.
Profile setup	The user finishing their profile: the CBI, the Comfort Zone and optional ceiling fine-tuning.
Complete Resident Profile	A profile whose CBI is complete; the resident brewing paths are unlocked.
Profile extras	Additive preferences, grind setting, Sub-Profiles, the BrewNow! Profile and workplace settings, created in the app.
Computed value / adjustment / effective value	For each ceiling: the learned value, the user's own setting if any, and the one the device uses.
Today's plan / Routine	A plan for one day, built on either surface and not saved; a saved plan kept for its saved days, managed in the app.
Planned bedtime / session bedtime	The bedtime entered with a plan; the bedtime confirmed at the device with each brew, which always rules.
Invite code	A one-time code, emailed to the invited address, that creates the account and claims the profile.
Visitor pass	A short-lived, machine-bound permission, granted from the app, to use the profile for one visit.
Standing pass	A saved consent for one machine, for a regular stop, kept in the user's account.
Visit token	The signed message the machine receives from the cloud, stating what a pass covers and when it ends.
Places list	The app's list of regular stops, any visit in progress, and visit history.
Caffevolve Away	Working name for the visitor pass concept.